NIHARIKA KANWAR

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Education

Michigan State University **Aug. 2023**
Bachelor of Science in Computer Science, business (Minor) *Lansing, Michigan*

Technical Skills

Languages: Python, SQL, C++, Java, JavaScript, HTML, CSS, Swift
Technologies: Docker, AWS, Firebase, Google Cloud, Linux, AArch64 Assembly, VS Code, WCAG 2.1, Digital Accessibility

Relevant Coursework

Data Structures and Algorithms, Object Oriented Programming, Computer Organization and Architecture, Database Systems, Discrete Mathematics, Matrix (Linear) Algebra, Web Development

Experience

MATASMA Digital Technologies | Data Science Intern July 2025 – August 2025
ARIMA, SARIMA, Pandas, NumPy, Python

- Leveraged data analytics skills to analyze retail sales dataset (Rossman sales) using Python (Pandas, NumPy) to identify seasonality patterns, holiday effects, and promotional impacts across 1,115 stores for business forecasting
- Engineered time-series features, including lag variables, rolling averages, and holiday flags, to improve predictive accuracy of forecasting models by 18% over baseline
- Developed and compared ARIMA, SARIMA, and Prophet models with walk-forward validation and hyperparameter tuning, achieving 87% accuracy (MAPE ~13%)
- Created dashboards using Matplotlib and Seaborn to visualize promotional impacts and seasonal trends, supporting business forecasting and scenario planning

MSU Computer Science Department | Undergraduate Learning Assistant Jan 2026 – Present
Assembly (AArch64/MIPS), C, Computer Architecture, Digital Logic, Debugging, ISA, Memory Hierarchy *Lansing, MI*

- Supported 180+ students in a systems-level computer architecture course during helproom sessions, debugging issues in assembly language, data representation, and instruction execution
- Explained Boolean algebra, combinational/sequential circuits, and ISA behavior, helping students connect hardware design principles to low-level software behavior

Human Buildings Systems Lab MSU | Undergraduate Research Assistant Python, October 2024 – December 2024
Genetic Algorithms, Optimization, Simulation, Data Analysis *Lansing, Mi*

- Collaborated with a PhD researcher to optimize electric vehicle charging station placement using Genetic Algorithms in Python, analyzing constraint satisfaction and sustainability metrics
- Debugged code errors, verified results, and compared model outputs against real-world infrastructure data

Projects

Lost&Found Dec 2025-Jan 2026
AWS Lambda (Python), API Gateway, DynamoDB (GSI, TTL), Amazon S3, IAM, CORS

- Designed a serverless lost-and-found platform for 30+ residence hall service desks using AWS (Lambda, API Gateway, DynamoDB, S3), enabling campus-wide photo uploads, search, and automatic scaling without server maintenance
- Built a REST API with DynamoDB GSIs for multi-attribute queries and implemented secure image storage with S3 (CORS, IAM least-privilege) and DynamoDB TTL for automated 90-day data retention

CharmBooth - Photo Booth Web App | **https://github.com/niharika-kanw/charmbooth** Jan 2026
React (CDN), HTML5 Canvas, getUserMedia, MediaRecorder, Tailwind CSS, JavaScript

- Built web camera app with real-time filters and animated particle overlays using WebRTC, Canvas API, and React, processing live video at 30fps
- Implemented particle system with multiple effect types (hearts, sparkles, bubbles) using object-oriented design and requestAnimationFrame for smooth animations
- Added 4-second video recording with MediaRecorder API and Web Share integration for easy clip export and sharing

Spartan Sync (Campus Event Platform) https://github.com/niharika-kanw/SpartanSync February 2026
React, Vite, TypeScript, Tailwind CSS, Supabas, ElevenLabs

- Built an event platform using React and Supabase with real-time RSVP synchronization via PostgreSQL subscriptions, participant caps, and event-specific comment threads
- Integrated ElevenLabs text-to-speech for accessible event narration and implemented Supabase Auth with domain-restricted email validation

Leadership / Extracurricular

Women in Engineering Mentor, AWS Cloud Club Captain @ MSU, AWS 10,000 AIDEAS current semi-finalist, Resident Assistant (1.5+ years), AWS Cloud Practitioner Certification, Business discovery program member, Digital Accessibility Assistant